nationals to own land; impose joint venture requirements on foreign investment in particular sectors to encourage technology transfer to local firms; and place restrictions and conditions on foreign investment to protect local firms from foreign competition. Absent relevant exceptions, such measures breach NT in the pre-establishment phase.[19] For instance, Singapore applies an additional buyer's stamp duty of 15 per cent to foreign nationals buying land.[20] As a result of the unqualified NT obligation in the Singapore–US FTA 2001, US citizens are entitled to the same treatment as Singaporeans, and, consequently, do not need to pay the higher rate of tax (unlike, for example, Chinese, Indonesian, or Malay citizens purchasing property in Singapore) (Vaughan 2013).

Pre-establishment NT and MFN obligations have played no role in investment treaty arbitration to date. Time will tell whether they acquire greater importance. Yet, even in the absence of claims brought to arbitration, the extension of MFN/NT obligations to the pre-establishment phase could have powerful behind-the-border effects in requiring host states to remove or amend discriminatory restrictions on foreign investment. We return to the issue of the impact of investment treaties on state decision-making in Chapters 5, 6, and 9—an important area for further research.

Absolute standards of protection

We now turn to four 'absolute' standards of protection: expropriation, FET, umbrella clauses, and free transfer of funds. These standards of protection are 'absolute' in the sense that they require host states to guarantee certain standards of treatment to foreign investors and investments covered by the treaty, regardless of how other investors and investment are treated. We focus on these four absolute standards, as they typically capture breaches of other substantive protections as well—such as the prohibition of arbitrary, unreasonable, or discriminatory measures, which overlaps significantly with FET and NT.

NO EXPROPRIATION WITHOUT COMPENSATION

Expropriation refers to state conduct that deprives investors of title to property, or otherwise substantially deprives investors of their investments.[21] Investment treaties allow states to expropriate foreign investments, but this right is subject to the payment of compensation (see again the discussion of

[19] Figure 8.4 shows exceptions to pre-establishment national treatment in selected US treaties.
[20] Inland Revenue Authority of Singapore (2016).
[21] We use the terms 'nationalization' and 'taking' as synonyms for 'expropriation'.

'liability rules' in Chapter 1). Recall from Chapter 2 that most investment treaties cover a broad range of investments, including intangible rights (e.g. intellectual property rights) and financial instruments (e.g. bonds and hedging contracts). As such, the application of investment treaties' expropriation provisions extends beyond the expropriation of physical assets such as factories and mines.

The legal principles governing expropriation of foreign investment are rooted in customary international law. Customary international law acknowledges a state's right to expropriate foreign-owned property subject to three conditions: the expropriation must (i) serve a public purpose; (ii) be carried out with due process of law; and, most importantly, (iii) be accompanied by 'full' compensation—a standard understood to refer to an investment's fair market value (see section on 'Compensation and Damages' in the following text). The duty to pay compensation is by far the most important condition in practice; although many expropriations comply with conditions (i) and (ii), investors frequently contend that host states fail to pay full compensation.

These standard conditions are also found in most investment treaties. For example, Article 5 of the 2008 UK Model Investment Treaty provides:

> Investments of nationals or companies of either Contracting party shall not be nationalized, expropriated or subjected to measures having effect equivalent to nationalization or expropriation in the territory of the other Contracting Party except for a public purpose related to the internal needs of that Party on a non-discriminatory basis and against prompt, adequate and effective compensation.

As with almost all investment treaties, Article 5 distinguishes between direct expropriations ('investments . . . shall not be nationalized [or] expropriated') and indirect expropriations ('or subjected to measures having effect equivalent to nationalization or expropriation'). Both are equally compensable.

Direct expropriation involves a transfer of title or physical seizure of an investment. The expropriation comes about in a single act, and is usually easy to identify. As direct expropriations without compensation are becoming rare (see Chapter 1), only about one in ten of the more than 700 known investment treaty arbitrations have alleged a violation of this provision (Table 4.2). Instead, the more important question in arbitral practice is how to identify *indirect* expropriation. Investors have alleged indirect expropriation in 45 per cent of all known investment treaty arbitrations (Table 4.2),[22] and the subject has proven highly controversial (Fortier and Drymer 2004; Ratner 2008, 229, 482–4; Schneiderman 2008, 75; Shirlow 2014). Several considerations are important for this distinction, which we review in the following text.

[22] The 2008 UK Model BIT, Article 5, for instance, describes indirect expropriation as 'measures having effect equivalent to nationalization or expropriation'.

As a starting point, it is important to clarify that mere reductions in the profitability of an investment are insufficient for an indirect expropriation to have occurred; what is needed is a loss of such magnitude that it very substantially lowers the value of an investment. For example, in one recent case, a tribunal concluded that significant reductions in the feed-in tariffs that Spain paid to foreign-owned solar plants did not amount to expropriation (*Charanne v. Spain* 2016).[23] Equally, even though Argentina's modification of gas distribution licenses in the early 2000s in the aftermath of its financial crisis had a negative impact on the businesses of foreign investors, a tribunal found that the impact was not severe enough, nor sufficiently permanent in duration, to amount to indirect expropriation (*LG&E v. Argentina* 2006, para. 200). Similarly, host states can significantly increase levels of taxation without the resulting tax burden amounting to an indirect expropriation (*Paushok v. Mongolia*, para. 332).

While arbitral practice unanimously supports the view that state conduct can only amount to indirect expropriation if it results in a substantial deprivation of the investor's investment, there is an ongoing controversy about how to draw the line between indirect expropriation and a state's non-compensable exercise of regulatory powers to meet public policy objectives. There are many cases in which state conduct has a substantial negative impact on an investor's business but in which the state asserts a justification for such interference on public policy grounds. Examples from previous arbitrations include a host state cancelling an investor's operating license citing the failure of the investor to meet mandatory environmental standards (e.g. *Tecmed v. Mexico* 2003), and a state banning the sale of an investor's product citing public health risks (e.g. *Chemtura v. Canada* 2010). Some tribunals have articulated implicit exceptions to the concept of indirect expropriation—namely, that non-discriminatory regulatory measures in pursuit of legitimate policies *never* amount to indirect expropriation regardless of the magnitude of loss, or interference with, an investment (e.g. *Methanex v. US* 2005, Part IV, Chapter D, para. 7).

More generally, there are two stylized approaches in determining whether a measure affecting a foreign investment amounts to indirect expropriation and, therefore, obliges the host state to compensate the investor. The first approach looks exclusively to the *effect* of the state conduct on the investment, including the degree of economic loss that the conduct causes to investors and the extent to which it interferes with the investor's legal rights to use and enjoy the property (Schreuer 2006; Hoffmann 2008). This view accepts that state conduct must have created a 'persistent or irreparable obstacle' to the operation of the investment (*Generation Ukraine* 2003, para. 20.32), but permits little or no

[23] At the time of writing, more than 30 similar arbitrations were pending against Spain concerning its modifications of feed-in tariffs for solar energy.

consideration of the state's regulatory purpose. This 'effects' approach was particularly influential in early investment treaty arbitral decisions. For example, in *Metalclad v. Mexico* (2000), the tribunal held that a local government's refusal to issue a construction permit to a landfill that had been approved by all other levels of the Mexican government amounted to an indirect expropriation. In the words of the tribunal, the local government's inaction was a 'covert or incidental interference with the use of property which has the effect of depriving the owner, in whole or in significant part, of the use or reasonably-to-be-expected economic benefit of property even if not necessarily to the obvious benefit of the host State' (para. 103).

A second approach adopts a narrower view of indirect expropriation, granting states greater latitude to regulate foreign investment for legitimate public purposes. Alongside the intensity of interference with the property of the investor, this approach considers the *purpose* and *characteristics* of host state measures. For example, in another case against Mexico, the tribunal considered Mexico's decision to discontinue its practice of granting tax rebates on cigarette exports to a foreign investor. In addition to noting that the measure had not deprived the investor of control over its investment, the tribunal observed that the change in practice affected *all* cigarette resellers and was justified by the purpose of discouraging tobacco smuggling (*Feldman v. Mexico* paras. 112–16, 136–7). Taken together, these factors meant that Mexico's measure did not amount to indirect expropriation.

The 'effects' approach to indirect expropriation has been especially controversial, as critics argue that it provides greater protection to foreign investors than is provided to private property rights in most developed legal systems (Montt 2009). In response to these, and other, criticisms, a newer generation of investment treaties define the concept of 'indirect expropriation' more precisely, thereby reducing arbitral tribunals' interpretative discretion (Nikièma 2012; Moloo and Jacinto 2013). The first and most influential clarification was a new annex to the 2004 US Model BIT, which clarifies that non-discriminatory regulatory measures will only amount to indirect expropriation 'in rare circumstances'.[24] It also directs arbitral tribunals to distinguish between indirect expropriation and legitimate, non-compensable regulation based on three factors:

(1) the degree of interference with property rights;
(2) the character of the government measure—that is, its purpose and context; and
(3) the interference of the measure with reasonable and investment-backed expectations—notably, expectations related to the use of the investment.

[24] Annex B on Expropriation, US Model BIT 2004. Canadian investment treaties follow a similar approach.

These three factors are adapted from the US Supreme Court's jurisprudence on indirect expropriation, notably the decision in *Penn Central Transportation Co. v. New York City' (1978)* (Sanders 2010). The US government's decision to define 'indirect expropriation' based on the way that concept is understood in US law followed a Congressional direction to the US Executive in 2002 that US investment treaties should not provide protection to foreign investment beyond that found in US law (Gagné and Morin 2006). The annex on indirect expropriation has been included in all subsequent US investment treaties. It has also diffused into the treaty practice of other states. Similar text clarifying the meaning of indirect expropriation is now found in investment treaties as diverse as the ASEAN Comprehensive Investment Agreement (ACIA), the Comprehensive Economic and Trade Agreement (CETA) between Canada and the EU, the China–Japan–Korea trilateral investment treaty, and the TPP.[25]

FAIR AND EQUITABLE TREATMENT

We turn now to the most important substantive protection in the investment treaty regime, FET. Consider the following sample clause:

> 'Each Contracting State shall in its territory in every case accord investments by investors of the other Contracting State fair and equitable treatment'.[26]

As with most FET clauses, this short and open-ended formulation provides broad interpretive discretion to investment tribunals. Whereas the outcome in early investment treaty arbitrations often turned on the application of indirect expropriation, the interpretation and application of FET provisions has been decisive in most disputes since 2000. (This is an important point for empirical scholarship on investment treaty arbitration, some of which still focuses primarily on indirect expropriation; e.g. Pelc 2016). Investors have invoked the FET standard to challenge the entire gamut of state conduct, including changes to legislation of general application, decisions of executive agencies of the host state specifically addressed to the investor in question, and the actions of the host state's judiciary.

One important point of contrast between FET and the concept of indirect expropriation is that host states may breach FET even if the impact of state conduct on the foreign investor falls short of a 'substantial deprivation' of an investment necessary for expropriation. Moreover, in contrast to NT and MFN, a host state's action may breach FET even if it is non-discriminatory. Investment tribunals 'regularly apply [FET] in a broad manner, using it as a yardstick for the conduct of the national legislator, of domestic administrations, and of domestic courts' (Schill 2009, 79). For this

[25] Annex 2, ASEAN Comprehensive Investment Agreement 2009; Annex 8-A, CETA 2016; protocol to the China–Japan–Korea trilateral investment treaty 2012; Annex 9-B, TPP 2016.

[26] Article 2(2), German Model BIT 2009.